AI Agents, MCP (basic introduction)

Week 1: Modern AI

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Session Topics

- ✔ What is an AI Agent?
- ✔ Components of an AI Agent
- ✔ How Does an AI Agent Work?
- ✔ What is MCP?
- ✔ Basic MCP Architecture
- ✔ MCP Tools & Their Role in AI Agents

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What are AI Agents?

- An AI Agent is a system that can perceive information, reason about a goal, make decisions, and take actions.
- Unlike a traditional chatbot, an agent can perform tasks instead of only generating responses.
- Agents can use LLMs, tools, memory, and external systems to accomplish tasks.
- An agent can break a complex goal into smaller steps and execute them.
- The level of autonomy depends on how the agent is designed and the permissions it has.

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Components of an AI Agent

An AI agent typically combines several components:

Goal: Defines what the agent needs to accomplish.

LLM / Reasoning: Interprets the task and determines the next steps.

Memory: Stores relevant information from previous interactions or actions.

Tools: Allow the agent to interact with external systems such as APIs, databases, search, or applications.

Planning: Breaks a complex task into smaller actions and determines their sequence.

Action: Executes the selected tool or operation.

Observation: Receives the result and uses it to decide what to do next.

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How Does an AI Agent Work?

An AI agent typically follows an iterative process.

1. **Receive Goal:** The user provides a task or objective.
2. **Understand:** The agent interprets the request and identifies what needs to be accomplished.
3. **Plan:** The agent determines the steps required to achieve the goal.
4. **Select Tool:** The agent chooses an appropriate tool or external system.
5. **Take Action:** The selected tool is executed.
6. **Observe Result:** The agent receives and evaluates the result.
7. **Continue or Finish:** The agent performs another action if necessary or provides the final result.

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What is MCP(Model Context Protocol)?

- MCP (Model Context Protocol) is an open standard that provides a common way for AI applications to connect with external data sources, tools, and services.
- It helps AI applications discover and use capabilities provided by external systems.
- MCP can connect AI systems to files, databases, APIs, business applications, and other tools.
- Instead of creating a different integration for every AI application and every tool, MCP provides a standardized communication approach.
- MCP is particularly useful for building tool-using AI agents and agentic workflows.

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Basic MCP Architecture

MCP uses a **client-server architecture**.

MCP Host

- The AI application that the user interacts with.
- Example: an AI assistant or agent application.

MCP Client

- The component that maintains the connection with an MCP server and communicates using the MCP protocol.

MCP Server

- Provides access to specific tools, resources, or capabilities.
- It acts as a bridge between the AI application and an external system.

External System

- The actual system being accessed.
- Examples: database, file system, API, business application, or search service.

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MCP Tools & AI Agents

MCP enables AI applications to interact with external systems through standardized tools and interfaces.

Common MCP Tool Capabilities

- **Information Retrieval** → Retrieve relevant information from connected data sources.
- **Database Operations** → Query and retrieve structured business data.
- **File & Document Access** → Read and process files from connected systems.
- **API Integration** → Interact with external applications and services.
- **Business Operations** → Execute authorized actions within connected business systems.

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